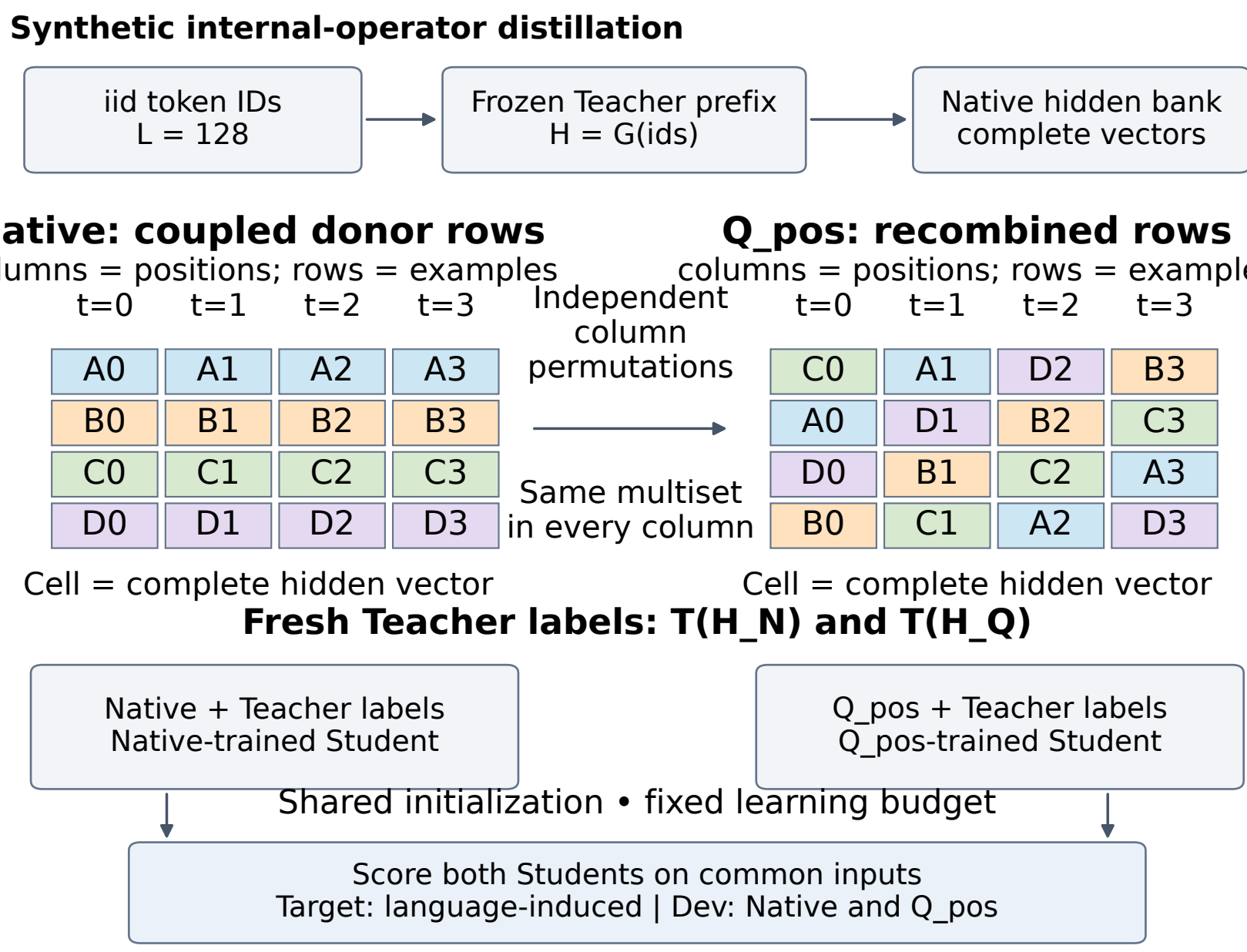
Synthetic internal-operator distillation
iid token IDs
L = 128
Frozen Teacher prefix
H = G(ids)
Native hidden bank
complete vectors
Native: coupled donor rows
columns = positions; rows = examples
t=0 t=1 t=2 t=3
A0 A1 A2 A3
B0 B1 B2 B3
C0 C1 C2 C3
D0 D1 D2 D3
Cell = complete hidden vector
Independent
column
permutations
Same multiset
in every column
Q_pos: recombined rows
columns = positions; rows = examples
t=0 t=1 t=2 t=3
C0 A1 D2 B3
A0 D1 B2 C3
D0 B1 C2 A3
B0 C1 A2 D3
Cell = complete hidden vector
Fresh Teacher labels: T(H_N) and T(H_Q)
Native + Teacher labels
Native-trained Student
Q_pos + Teacher labels
Q_pos-trained Student
Shared initialization • fixed learning budget
Score both Students on common inputs
Target: language-induced | Dev: Native and Q_pos